[PLACEHOLDER] Developement of a Segmented Communication Protocol for Sensor Data Transmission Using LoRa© Technology

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***Abstract*—This paper presents a lightweight, hardware-agnostic protocol for segmenting and reliably transmitting large data payloads over Long Range (LoRa©) physical links. The protocol is designed to maximize throughput over long-range, point-to-point LoRa links, enabling fast transfer of large structured payloads (for example, images or telemetry logs). Packets carry a compact binary header and per-packet CRC. We describe the packet layout, reassembly rules, and a proof-of-concept implementation using Semtech SX126x-class transceivers. Experimental results demonstrate ... [PLACEHOLDER]**

***Index Terms*—**

I. Introduction

LOW-POWER, low-cost and long-range wireless communication technologies are becoming increasingly important in the context of the Internet of Things (IoT). Applications such as environmental monitoring, precision agriculture or Unmanned Aerial Vehicles (UAVs), require reliable data transmission over extended distances with strict power and hardware constraints [1].

A range of different communication standards, including ZigBee, Sigfox and LoRa© have been adopted to address these needs, and LoRa©, in particular, has emerged as the leading choice for low-power and long-range deployments thanks to its robust modulation technique and open network architecture. However, conventional solutions based on LoRa© (particularly those following the Long Range Wide Area Network (LoRaWAN©) specifications) are optimized for small, non-frequent payloads.

This paper presents the design and implementation of a custom, hardware-agnostic communication protocol for LoRa©, engineered to enable the transmission of segmented large data payloads over long distances without relying on higher-layer middleware (e.g., LoRaWAN©). The protocol focuses on compact headers, per-packet integrity checks, and a small set of reliability primitives designed to maximize throughput in point-to-point LoRa links while keeping airtime overhead minimal. The proposed system has been deployed on a sounding rocket built by the Aurora Rocketry team in the University of Bologna, demonstrating its applicability to highly dynamic systems, that require continuous transmission of large amount of telemetry data with high reliability and low power consumption in a point-to-point configuration. This approach is generalizable to a wide range of IoT applications requiring extended data capacity over extended distances and high power efficiency, and possibly to multi-agent systems that require the transmission of structured data between the agents.

The aim of this work is to produce a general-purpose lightweight framework for the transmission of large data payload over LoRa©

II. State of The Art

III. Proposed System

The protocol uses a compact, fixed-layout binary header designed to minimize airtime while still providing the information necessary for reassembly and selective retransmission. The implementation described in this paper uses the following header fields (all multi-byte fields are encoded little-endian):

- **protocolVersion (1 byte)**: Format version for forward/backward compatibility.
- **flags (1 byte)**: Bitfield where bit0=Start-Of-Message (SOM), bit1=End-Of-Message (EOM), bits2–7 reserved for future use (e.g., ACK required, etc.).
- **messageId (2 bytes)**: Short identifier that groups fragments belonging to the same logical message.
- **totalChunks (1 byte)**: Number of fragments composing the message.
- **chunkIndex (1 byte)**: Zero-based index of this fragment within the message (0..totalChunks-1).
- **payloadSize (1 byte)**: Number of valid payload bytes carried by this packet (0..255). Use uint16 if larger per-chunk MTU is required.

After the payload, a 16-bit CRC (CRC-16, recommended CRC-16-CCITT) is appended and covers the header and payload; the CRC field itself is excluded from the CRC calculation.

This layout yields an 8-byte header (excluding the trailing CRC). The design is deliberately small: for typical LoRa© MTUs the per-packet overhead is low while the messageId and flags provide the primitives needed for grouping and duplicate detection.

IV. Preliminary Proof of Concept

An initial prototype used commercial Ebyte E220-series modules and validated the segmentation concept by splitting large telemetry payloads into fixed-size chunks. That prototype demonstrated feasibility during flight tests but highlighted three important limitations: (1) the modules exposed a proprietary network layer and did not allow full control of the physical/link layer parameters, (2) repeating the full header in every chunk created measurable airtime overhead, and (3) there was no lightweight acknowledgment or retransmission mechanism.

To address these shortcomings we implemented a hardware-agnostic proof-of-concept on Semtech SX126x-class transceivers (demonstrated on a Heltec WiFi LoRa 32 V3 development board). The new PoC keeps the protocol independent of vendor-specific APIs and focuses on the primitives described above: an 8-byte header (protocol version, flags, messageId, totalChunks, chunkIndex, payloadSize), a per-packet CRC-16 covering header+payload, and a small set of reliability helpers. Receivers reassemble messages by matching the messageId and collecting fragments indexed 0..totalChunks-1. Duplicate fragments are ignored by a per-message received-bitmap.

V. Hardware Level Implementation

A. Implementation

B. Testing

VI. Conclusion

The conclusion goes here.

Appendix A
Title of Appendix One

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Appendix B

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Acknowledgment

The authors would like to thank...

References

[1] M. Radeta, J. Pestana, P. Abreu, R. Freitas, F. Silva, D. Vieira, R. Prieto, M. Fernandez, F. Alves, T. Dellinger, S. Neves, and E. Delory, “Triton—open telemetry and location estimation for marine monitoring based on iot and lora,” *IEEE Journal of Oceanic Engineering*, vol. 50, no. 2, pp. 1244–1258, 2025.

Alessandro Monticelli Biography text here.